The Impact of Metformin, Oral Contraceptives, and Lifestyle Modification on Polycystic Ovary Syndrome in Obese Adolescent Women in Two Randomized, Placebo-Controlled Clinical Trials

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Context: Polycystic ovary syndrome (PCOS) presents in adolescence, and obesity is a common finding. The benefits and risks of alternate approaches to the management of PCOS in obese adolescent women are not clear.

Objective: We investigated the effects of metformin, oral contraceptives (OCs), and/or lifestyle modification in obese adolescent women with PCOS.

Design: Two small, randomized, placebo-controlled clinical trials were performed.

Patients and Participants: A total of 79 obese adolescent women with PCOS participated.

Interventions: In the single treatment trial, subjects were randomized to metformin, placebo, a lifestyle modification program, or OC. In the combined treatment trial, all subjects received lifestyle modification and OC and were randomized to metformin or placebo.

Main Outcome Measures: Serum concentrations of androgens and lipids were measured.

Results: Lifestyle modification alone resulted in a 59% reduction in free androgen index with a 122% increase in SHBG. OC resulted in a significant decrease in total testosterone (44%) and free androgen index (86%) but also resulted in an increase in C-reactive protein (39.7%) and cholesterol (14%). The combination of lifestyle modification, OC, and metformin resulted in a 55% decrease in total testosterone, as compared to 33% with combined treatment and placebo, a 4% reduction in waist circumference, and a significant increase in HDL (46%).

Conclusions: In these preliminary trials, both lifestyle modification and OCs significantly reduce androgens and increase SHBG in obese adolescents with PCOS. Metformin, in combination with lifestyle modification and OC, reduces central adiposity, reduces total testosterone, and increases HDL, but does not enhance overall weight reduction. ***(J Clin Endocrinol Metab* 93: 4299–4306, 2008)**

Polycystic ovary syndrome (PCOS) is a heterogeneous condition characterized by chronic anovulation and androgen excess that occurs in 4–8% of unselected women (1). Signs and symptoms of PCOS typically appear at the time of puberty (2). At least 50% of women with PCOS are obese, resulting in a more severe clinical picture (3). Obesity among adolescents is prevalent (4), with overrepresentation in those with evidence of hyperandrogenism and irregular periods

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Abbreviations: AUC, Area under the curve; BMI, body mass index; CH, cholesterol; CRP, C-reactive protein; cv, coefficients of variation; FAI, free androgen index; HDL, high-density lipoprotein; Hs-CRP, highly sensitive CRP; IGT, impaired glucose tolerance; LDL, low-density lipoprotein; OC, oral contraceptive; OGTT, oral glucose tolerance test; PAI-1, plasminogen activator inhibitor-1; TG, high-density lipoprotein; UPDG, urinary pregnanediol glucuronide.

(5, 6), suggesting an association of obesity and PCOS at an early age.

Recent data have drawn attention to the long-term metabolic risks of PCOS (7–11). Insulin resistance plays a critical role in the pathophysiology of PCOS (12) and is thought to be the metabolic abnormality most closely linked to an increased risk of diabetes and heart disease (13, 14). It is also likely that the precursors to cardiovascular disease or diabetes are already present in the adolescent with PCOS, and there is a need to address these factors as well as the endocrine dysfunction. The best management for the long-term treatment of PCOS is not known. Treatment options include oral contraceptives (OCs), lifestyle modification, and insulin sensitizers. There are few head-to-head trials to distinguish between first-line therapies in the adolescent.

Traditional treatment with OCs is associated with reduction in androgens and improvements in menstrual cycles in adolescents with PCOS (15). Few studies have been reported in obese adolescents (16). Whether OCs worsen glucose tolerance or metabolic dysfunction in PCOS is controversial (17–20). Nonetheless, OCs remain a first-line therapy choice for the adolescent with PCOS.

Lifestyle modification has been shown to be effective in the restoration of ovulation (21–25) but has not been studied well in obese adolescents with PCOS. There is growing evidence that obese adolescents with PCOS are at increased risk for metabolic consequences (26) and that treatment of obesity at this critical developmental juncture may provide long-term health benefits (26, 27).

Metformin, which may have peripheral insulin-sensitizing effects (28), has been shown to exert several beneficial effects in trials of adult women with PCOS (29). Although a recent large randomized trial did not show any fertility advantage to the use of metformin (30), other studies have demonstrated metabolic benefits (31, 32). Whether these benefits can be extrapolated to adolescent women is unclear.

The aims of these two small randomized trials were: 1) to gain initial information on the practical concerns of recruiting, retaining, and randomizing obese adolescent women with PCOS to studies involving lifestyle management; and 2) to assess the metabolic and endocrine response to the three most common treatments: lifestyle management, OCs, and metformin, either alone or in combination.

Subjects and Methods

Study designs

The initial trial, called the "single treatment trial," was designed to estimate the effects of three treatments individually—metformin, lifestyle management, and OCs. This was a randomized, placebo-controlled trial in which each of the three treatments, plus placebo, were randomly assigned. Given the paucity of information regarding the effectiveness of lifestyle management studies in the population of obese adolescent women with PCOS, attention to the acceptance and success of the lifestyle program was of primary interest.

The "combined treatment trial" was designed to allow for improvements in the lifestyle program, after review of the single treatment trial, and to assess combination therapy of metformin plus OCs on markers of cardiovascular risk. This was a randomized, placebo-controlled trial in which all participants received lifestyle modification and OCs. Although the main study aim was to assess success in recruiting and retaining adolescent subjects to a lifestyle modification program, it was also intended to provide initial assessment of impact on lipid parameters using combination therapy with an OC. Although it is not feasible to use cardiovascular events as outcome measures in adolescents, surrogate markers such as the triglyceride (TG)/high-density lipoprotein (HDL) ratio may provide a measure of cardiovascular risk (33). Subjects were randomized to one of two treatment arms: metformin or placebo.

The lifestyle modification program was restructured in the combined treatment trial to include a rolling group admission and support groups. A lecture series was provided during both trials that included standard nutritional lectures as well as reading food labels, reading a menu, portion sizing, and hands-on kitchen training. Additionally, more behavioral support was added including peer support and electronic communications. Structured group exercise was included weekly. Goals of therapy were based on initial caloric intake by diet record aiming for a 500 kcal/d deficit. Exercise was recommended to include 30 min/d of moderate to intense activity.

Subjects

A total of 79 subjects were randomized in the two trials. All subjects were postmenarchal adolescent women between the ages of 12 and 18 yr with body mass index (BMI) above the 95th percentile and evidence of menstrual irregularity (fewer than eight menses in the preceding year) and clinical or biochemical evidence of hyperandrogenism. Although a transabdominal ultrasound was performed on all randomized subjects, it was not used as a criterion for the diagnosis of PCOS.

Subjects were recruited by direct advertisement in the community or by referral by physicians. All subjects were studied at the University of Rochester Medical Center in the Division of Reproductive Endocrinology. The metabolic and hormonal studies were performed at the General Clinical Research Center at the University of Rochester Medical Center. Both studies were approved by the Research Subjects Review Board at the University of Rochester, and all subjects provided signed consent if they were 18 yr of age or older or gave assent with signed consent by parent or legal guardian.

For the single treatment trial, a total of 366 subjects were screened by telephone survey, and 70 presented for evaluation between August 2002 and September 2003. Of these, 27 either failed to meet criteria for PCOS or did not wish to be randomized. In total, 43 subjects were randomized. All subjects met criteria for PCOS based on menstrual irregularity and androgen excess including Ferriman-Gallwey score above 7, elevated androgens after excluding other causes of hyperandrogenism, and menstrual irregularity. All were otherwise healthy. All were studied off any hormonal therapy or insulin sensitizers for at least 2 months, without regard to the time in the menstrual cycle.

Recruitment for the combination treatment trial began in May 2006 and concluded in July 2007. Similar eligibility requirements applied to this study, and subjects were again recruited directly from the community or local physician offices. A total of 102 subjects were screened by telephone, and 41 presented for hormonal evaluation. Thirty-six met criteria and agreed to randomization. All were studied randomly at baseline off of any treatment for at least 2 months without regard to time in menstrual cycle. Subjects were otherwise healthy, and all were nonsmokers. No subject in the single treatment trial was also included in the combination treatment trial.

Study protocol: single treatment trial

Subjects were randomized to one of four arms by random number assignment: placebo, metformin, OC, or lifestyle management. Assignment to metformin or placebo was blinded to subject and investigator. Metformin and placebo capsules were prepared by the Investigational Drug Pharmacy service at the University of Rochester using a commercially available powdered form of metformin or a lactose powder for the placebo, dispensed into identical capsules.

A clinical assessment was performed at baseline and at 24 wk. Total testosterone, SHBG, and lipids including total cholesterol (CH), low-density lipoprotein (LDL)-CH, HDL-CH, and TG were drawn at baseline and 24 wk. Plasminogen activator inhibitor-1 (PAI-1) and highly sensitive (Hs) C-reactive protein (CRP) were also measured at baseline and 24 wk. Testing for glucose tolerance by an oral glucose tolerance test (OGTT) was performed using a 75-g 2-h glucose challenge. Dual-energy x-ray absorptiometry (Hologic software version 12.6.1; Hologic, Inc., Bedford, MA) was used to assess percentage body fat. A computed tomography scan of the abdomen at the level of L4-L5 was used to assess intraabdominal body fat (sc and visceral fat). Subjects were seen monthly for collection of menstrual data and pill checks. Urine for pregnanediol glucuronide (UPDG) assay was collected weekly by the subjects and stored at home in the freezer until the monthly visit.

Metformin was given at a dose of 1700 mg/d in divided doses of 850 mg, starting as single doses of 425 mg and building gradually to two capsules twice a day. Placebo administration was matched to metformin in identical dosing. An OC was given with a formulation of 30 μg of ethinyl estradiol and 0.15 mg desogestrel (Desogen; Organon Pharmaceuticals, Roseland, NJ) and were taken in a cyclic fashion.

Those patients randomized to receive the lifestyle modification program were assigned in a closed group format for intervention, with five to six subjects per group. Subjects and one adult family member (parent or guardian) participated in a series of classes for training in diet, exercise, and behavior modification skills. Frequent contact with participants and a combination of structured lectures and flexible personal strategies with an emphasis on self-esteem and social support was used. The format includes a 16-session core curriculum taught in separate adolescent and parent groups over the 24 wk, interspersed with individual appointments. Exercise was not monitored. Subjects who were not assigned to lifestyle treatment received standard office advice on nutrition and exercise for healthy living and were seen monthly.

Study protocol: combination treatment trial

In this protocol all subjects were assigned to lifestyle intervention and placed on an OC containing 30 μg of ethinyl estradiol and 3.0 mg drosperinone (Yasmin, Bayer Schering Pharma, Berlin, Germany). Subjects were assigned by a random number table to metformin or placebo. Metformin and placebo were provided by the Investigational Drug Pharmacy service at the University of Rochester. A clinical assessment was performed at baseline and at 24 wk. Total testosterone, SHBG, Hs-CRP and lipids including total CH, LDL-CH, HDL-CH and TG were drawn at baseline and 24 wk. Testing for glucose tolerance by OGTT was performed using a 75-g 2-h glucose challenge. Dual-energy x-ray absorptiometry was used to assess percentage body fat. All subjects had a transabdominal ultrasound to assess ovarian volume. Subjects were seen weekly for the lifestyle program, with pill counts and menstrual data collected monthly. Metformin or placebo was given divided into four doses with a gradual build up at study start. Metformin dose for this study was 2000 mg/d. After randomization, all subjects were asked to wear an ActiGraph monitor for 7 d. This is a lightweight device worn on the waist to monitor activity levels and sleep patterns. This was repeated at study conclusion.

All subjects received a lifestyle intervention program that involved each subject and one adult family member (parent or guardian) in a series of classes for training in diet, exercise, and behavior modification skills in an open group format with rolling admission. Frequent contact with participants and a combination of structured lectures and flexible personal strategies emphasized self-esteem and social support. The format included a core curriculum taught in separate adolescent and parent groups repeated over the 24 wk, interspersed with individual appointments. Exercise was monitored in a group setting on a weekly basis. Contact between subjects was encouraged outside the weekly setting with electronic communications such as internet chat groups to support continued lifestyle changes.

Assays

After sample collection, sera was extracted and stored at −80 C. Pre- and posttreatment assays were run simultaneously for each subject. Hs-CRP and SHBG were measured by the Immulite system (Diagnostic Products Corp., Los Angeles, CA), interassay coefficients of variation (cv) 6.7 and 8.7%, respectively. Insulin was measured by ^{125}I immunoradiometric assay (LINCO Research, St. Charles, MO), cv 4.7 and 3.3%. Testosterone was measured by RIA (Diagnostic Systems Laboratories, Webster, TX), cv 8.5%. PAI-1 was measured by two-site ELISA, cv 3.5 and 10%, normal range 5–66 ng/ml (34). Glucose was measured by YSI select analyzer (YSI Inc., Yellow Springs, OH). Total CH, HDL-CH, and TGs are measured by dry slide enzymatic colorimetric assay (Vitros products; Johnson & Johnson, New Brunswick, NJ). HDL was separated by precipitation of LDL and very low-density lipoprotein using dextran sulfate and magnesium chloride and was removed by centrifugation. LDL was assayed by enzymatic CH assay (Sigma Diagnostics, St. Louis, MO) after precipitation of very low-density lipoprotein and HDL. The free androgen index (FAI) was calculated by the equation: FAI = (testosterone/SHBG) × 100. The area under the curve (AUC) for glucose and insulin during the OGTT were calculated using the trapezoid method. The details of the assay for UPDG were previously reported (23).

Statistics

Results are expressed as mean ± SD. Statistical significance was noted for P values of ≤ 0.05. SPSS statistical package (SPSS Inc., Chicago, IL) was used for analysis. Differences between groups were compared using t tests for two samples or ANOVA for more than two groups for normally distributed variables. Within-group testing was by paired t test. Nonparametric testing was used for variables not normally distributed.

Results

Single treatment trial

Baseline characteristics

The baseline characteristics of all randomized subjects are presented in Table 1. There were no significant differences between groups. Impaired fasting glycemia, defined as fasting glucose greater than 100 mg/dl, or impaired glucose tolerance (IGT) with a 2-h glucose greater than 140 mg/dl was seen in 25% of subjects at baseline (four fasting and six IGT). One subject had 2-h glucose greater than 200 mg/dl, consistent with diabetes. The ethnicity of the subjects included 70% Caucasian, 19% African-American, 9% Hispanic, and 2% Asian.

Intervention results

Nine subjects did not complete the trial. Baseline characteristics are shown in Table 2. One subject from the placebo group and one from the OC group were lost to follow-up. In the metformin group, two subjects were lost to follow-up, and two left the study for personal reasons. In the lifestyle treatment group, three subjects left the study due to time commitment. Of those completing the lifestyle intervention, four subjects attended less than 50% of the lifestyle sessions and did not demonstrate weight reduction. The remaining subjects attended at least 75% of sessions and met lifestyle goals.

SHBG was significantly improved in the OC and lifestyle groups (Table 3). Total testosterone was decreased in the OC group, and FAI decreased in both the OC and lifestyle groups. Total CH, LDL-CH, and HDL-CH increased significantly in the OC group but did not change in the other groups, whereas PAI-1 declined significantly in the OC and lifestyle groups. Figures 1 and 2 demonstrate the change in SHBG and PAI-1, respectively,

TABLE 1. Baseline characteristics of subjects in single treatment trial

Variable	MET (n = 10)	PL (n = 11)	OC (n = 11)	LS (n = 11)	Normative values
Age (yr)	16 ± 1.7	15.4 ± 1.7	15.4 ± 1.4	15.4 ± 1.2	
BMI (kg/m^2)	34.3 ± 6.5	36.1 ± 7.5	37.8 ± 5.1	37.8 ± 8.2	<25
Waist (cm)	102.9 ± 10.8	108.7 ± 20.2	109.4 ± 14.1	113.5 ± 14.4	<88
Total T (ng/dl)	51.3 ± 13.0	61.2 ± 30.5	63.0 ± 23.0	63.9 ± 29.6	<40
SHBG (nmol/liter)	23.5 ± 19.1	17.7 ± 7.4	18.0 ± 11.7	14.6 ± 12.6	20–114
FAI	10.8 ± 5.8	14.8 ± 8.9	17.9 ± 13.3	21.7 ± 15.5	<4.5
FG score	7.8 ± 2.6	12.5 ± 5.3	9.8 ± 3.5	9.2 ± 1.9	<7
TCH (mg/dl)	161.5 ± 36.9	165.4 ± 24.7	168.1 ± 23.4	160.4 ± 27.4	<200
HDL (mg/dl)	40 ± 12.4	40 ± 5.4	36.9 ± 4.2	38.7 ± 9.4	>50
LDL (mg/dl)	109.7 ± 30.3	115.7 ± 21.2	120 ± 21.1	110.7 ± 25.5	<100
TG (mg/dl)	104.5 ± 47.4	89.8 ± 11.3	89.1 ± 7.5	89.2 ± 15.7	<150
Fasting ins (IU/ml)	20.8 ± 9.6	27.1 ± 17.2	23.3 ± 7.6	30.1 ± 16.9	<27
Fasting glu (mg/dl)	90.0 ± 7.6	89.1 ± 7.5	89.8 ± 11.2	89.2 ± 15.7	<100
SBP (mm Hg)	119 ± 11.0	114 ± 11.2	114.9 ± 18.3	112.3 ± 8.4	<130
DBP (mm Hg)	68.6 ± 12.9	64.6 ± 6.6	66.6 ± 7.8	68.6 ± 8.0	<85
CRP (mg/liter)	3.9 ± 3.2	4.3 ± 3.0	7.4 ± 6.1	5.5 ± 5.4	<3.0
PAI-1 (ng/ml)	48.8 ± 38.6	46.7 ± 32.7	46.5 ± 30.3	68.7 ± 34.5	2–47
AUC glu	22,333 ± 4,454	18,822 ± 2,563	19,687 ± 2,442	18,114 ± 2,772	
AUC ins	13,720 ± 8,184	33,352 ± 25,186	25,237 ± 11,219	29,192 ± 18,861	

MET, Metformin; PL, placebo; LS, lifestyle; T, testosterone; FG, Ferriman-Gallwey; TCH, total CH; SBP, systolic blood pressure; DBP, diastolic blood pressure; glu, glucose; ins, insulin.

by weight loss in the lifestyle treatment arm. Those who lost weight demonstrated a significantly greater increase in SHBG compared with those who did not. The decrease in PAI-1 was also greater and trended toward significance. Hs-CRP increased significantly in the OC group. TG decreased in the metformin group. Metformin alone reduced the TG/HDL ratio (−0.86; 95% confidence interval, −0.15 to −1.57), whereas there was a nonsignificant reduction in the OC group (mean change, −0.45; 95% confidence interval, −0.19 to 1.09). OGTT at the conclusion demonstrated new onset IGT in one subject in the metformin group, two in the placebo group, three in the OC group, and none in the lifestyle group (not significant).

Visceral fat increased significantly in the placebo group (+9.6 cm; $P = 0.04$) but did not change significantly in the other groups. Menstrual cycles did not differ between those not receiving OC; the metformin group reported an average of 3.2, lifestyle 2.3, and placebo 2.5 cycles per 24 wk. When ovulation was measured by weekly UPDG in the groups not receiving OC, 75% of the identified cycles were ovulatory in the metformin group, 60% in the lifestyle group, and 50% in the placebo group.

Combination treatment trial

Baseline characteristics

Baseline characteristics of the randomized groups in the combined treatment trial are shown in Table 4. Impaired fasting glycemia or glucose tolerance in all subjects was noted in 21.9% (two with impaired fasting and six with IGT). The ethnicity of the subjects included 75% Caucasian, 16% African-American, 6% Hispanic, and 3% Asian.

Intervention results

Four subjects did not complete the trial (Table 2). Two left from the metformin group and two from the placebo group. One subject in each of the groups stopped metformin or placebo due to gastrointestinal complaints, but each continued the OC and lifestyle program. Additionally, two subjects reduced the number of pills they took regularly in the metformin group but took at least 1000 mg/d. The majority of subjects (94%) completing the trial attended at least 75% of the lifestyle sessions. The remainder completed more than 50%.

In the placebo group, 12 of 16 subjects reduced their BMI, compared with 14 of 16 in the metformin group (not significant). A total of 12.5% had impaired fasting glycemia or IGT in the placebo group at baseline and 18.7% at conclusion, and 31% did so at baseline and conclusion in the metformin group. One subject in each group had fasting glucose greater than 126 mg/dl at baseline, and both of these subjects reverted to normal glucose at conclusion.

Table 5 reports the baseline and 24-wk assessment. BMI was reduced in both groups but did not differ between groups. Subjects in the metformin group significantly reduced waist circumference. Both groups reduced total testosterone, but those in the

TABLE 2. Baseline characteristics of subjects not completing trials

	Single treatment trial (n = 9)	Combined treatment trial (n = 4)
Age (yr)	15.9 (1.9)	14.5 (2.1)
BMI (kg/m^2)	38.5 (9.3)	36.7 (4.4)
Total testosterone (ng/dl)	60.2 (28)	
SHBG (nmol/liter)	26.2 (21.1)	9.9 (6)
Total CH (mg/dl)	169 (38.7)	131 (31.4)
HDL (mg/dl)	37.5 (7.7)	32 (3.2)
LDL (mg/dl)	119.5 (31)	89.5 (28.5)
TG (mg/dl)	112.6 (52.7)	52 (24.7)
Fasting glucose (mg/dl)	95.5 (12.4)	82 (4.2)
Ferriman-Gallwey score	9.5 (5.6)	7.5 (3.5)
Waist circumference (cm)	116.3 (17.8)	107.8 (5)

Data represent mean (SD).

TABLE 3. Baseline and 24-wk measures for subjects completing single treatment trial

	MET (n = 6)		PL (n = 10)		OC (n = 10)		LS (n = 8)	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
BMI (kg/m^2)	35.0 ± 8.2	35.7 ± 8.6	34.9 ± 6.7	35.5 ± 6.8	37.8 ± 5.3	36.4 ± 5.4[a]	36.0 ± 6.2	34.9 ± 7.0
Waist (cm)	100.5 ± 11.8	105.3 ± 13.9	104.7 ± 15.9	105.3 ± 18.6	108.8 ± 14.7	108.3 ± 16.1	110.9 ± 14.2	109.9 ± 17.3
Total T (ng/dl)	47.8 ± 15.1	49.7 ± 31.1	64.4 ± 31.4	71.6 ± 33.8	62.0 ± 24	34.5 ± 28.6[b]	60.7 ± 23.6	64.5 ± 30.2
SHBG (nmol/liter)	18.6 ± 8.9	21.1 ± 8.4	17.1 ± 7.5	19.1 ± 9.4	16.1 ± 10.3	93.2 ± 66.5[b]	14.4 ± 14.9	32.0 ± 21.7[a]
FAI	10.8 ± 5.6	10.9 ± 7.9	15.6 ± 8.9	16.8 ± 11.2	19.1 ± 13.5	2.4 ± 2.5[b]	23.2 ± 16.6	9.5 ± 5.3[a]
FG score	8.3 ± 3.1	8.2 ± 3.4	11.6 ± 4.5	11.6 ± 4.9	10.2 ± 3.5	8.6 ± 2.1	9.1 ± 1.5	8.2 ± 2.0
Total CH (mg/dl)	152 ± 22.9	145.3 ± 25	167 ± 25.5	157 ± 53.2	165 ± 22.5	188.6 ± 20.7[b]	158.5 ± 29	156.2 ± 31
HDL (mg/dl)	40.2 ± 14.9	43.5 ± 19	40.7 ± 5.1	43.6 ± 8.9	36.2 ± 3.7	47.6 ± 9.9[b,d]	40.9 ± 10.3	40.4 ± 7.6
LDL (mg/dl)	100.3 ± 19.2	92.0 ± 15.5	117.0 ± 22.0	114 ± 27.1	117.8 ± 20.5	128.6 ± 37.5[a,c]	107.6 ± 26.5	101.2 ± 32.3
TG (mg/dl)	94.8 ± 27.1	71.3 ± 21.1[a]	93.7 ± 30.6	87.1 ± 25.1	91.5 ± 41.2	96.1 ± 41.1	93.9 ± 31.7	109.6 ± 67.9
FI (IU/ml)	20.8 ± 10.9	19.8 ± 10.4	26.4 ± 17.9	29.1 ± 24.5	24.1 ± 7.6	20.7 ± 10.6	27.7 ± 16.4	22.0 ± 10.5
AUC glu	23,404 ± 5,329	19,373 ± 3,553	19,004 ± 2,625	18,721 ± 2,285	20,056 ± 2,228	19,567 ± 2,735	18,440 ± 3,178	17,843 ± 4,229
AUC ins	11,902 ± 3,755	12,661 ± 8,041	33,748 ± 26,512	32,538 ± 27,386	26,507 ± 10,961	18,580 ± 9,499[b]	32,119 ± 21,400	20,726 ± 16,153
FBS (mg/dl)	91.6 ± 7.9	84.9 ± 12.7[a]	87.6 ± 9.0	86.5 ± 5.4	89.7 ± 7.6	82.8 ± 9.8	81.4 ± 5.4	81.8 ± 9.1
SBP (mm Hg)	121.7 ± 13.8	122.7 ± 8.9	114.4 ± 11.7	117.5 ± 12.9	116.4 ± 18.6	124.1 ± 14.5	111 ± 9.7	119 ± 14.5
DBP (mm Hg)	74 ± 12.9	65 ± 8.6	65.3 ± 6.6	63.3 ± 4.7	67.4 ± 7.8	70.1 ± 8.3	70 ± 8.9	60.7 ± 9.4[a]
CRP (mg/liter)	3.6 ± 2.7	2.8 ± 2.0	4.28 ± 3.1	4.2 ± 2.8	6.8 ± 6.1	9.5 ± 7.4[a]	4.5 ± 3.8	3.8 ± 3.6
PAI-1	49.7 ± 37.9	45.4 ± 32.2	40.3 ± 26.1	48.0 ± 45.9	46.9 ± 31.9	29.5 ± 20.6[a]	72.1 ± 38.4	45.0 ± 25.6[a]

MET, Metformin; PL, placebo; LS, lifestyle; T, testosterone; FG, Ferriman-Gallwey; SBP, systolic blood pressure; DBP, diastolic blood pressure; FBS, fasting blood sugar; FI, fasting insulin; glu, glucose; ins, insulin.

[a] *$P < 0.05$ compared to baseline.*

[b] $P < 0.01$ compared to baseline.

[c] $P < 0.05$ compared to LS.

[d] $P < 0.05$ compared to LS, PL, and MET.

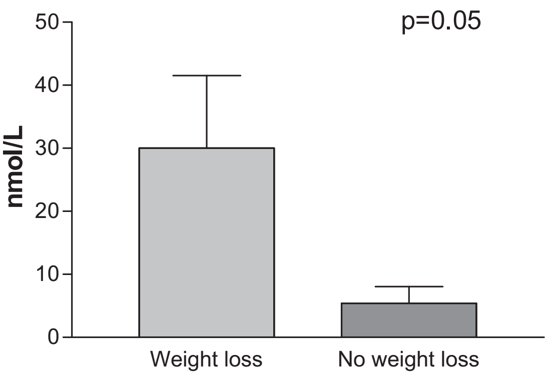

FIG. 1. A 24-wk change in SHBG for those completing the lifestyle treatment arm of the single treatment study separated by weight loss (n = 4 in each arm).

metformin group had significantly greater suppression. Both groups increased SHBG equally and had similar suppression of FAI. Clinically, both groups had significant suppression of Ferriman-Gallwey score. Both groups demonstrated increased CH, however the metformin group demonstrated significantly greater increases in HDL. Neither group demonstrated a change in TG.

Discussion

There is little information to guide clinicians in the management of the adolescent with PCOS, particularly in the presence of obesity (35). Moreover, the incidence of type 2 diabetes is on the rise in adolescence (36). Studies suggest a significant risk for lifetime obesity if one is obese as an adolescent (37). Obesity in adolescence predisposes for adult heart disease and is also associated with decreased fecundity (38, 39). To date there are few studies that assess the question of first-line management of PCOS in obese adolescent women. The data we present here are preliminary in nature because larger randomized trials are clearly needed. Nonetheless, the two small clinical trials reported here, the second building on the first, are presented to provide initial data on the most common management options that are offered to these adolescent women. OCs are a mainstay of therapy in PCOS, and studies in lean adolescent women with PCOS demonstrate worsening of some metabolic factors with the use of OC (18, 19). However, these abnormalities were found to be improved with combination metformin and/or antiandrogens in a normal-weight adolescent population (40). In our single treatment trial, a significant reduction was seen in androgens in the OC group. However, our results raise concerns related to the increases in CRP and increases in LDL-CH in the OC group, raising the possibility of inflammatory and metabolic disturbance with OC use in this population. Others have demonstrated an increase in Hs-CRP in OC *vs.* non-OC users (41), consistent with a low-grade inflammatory process.

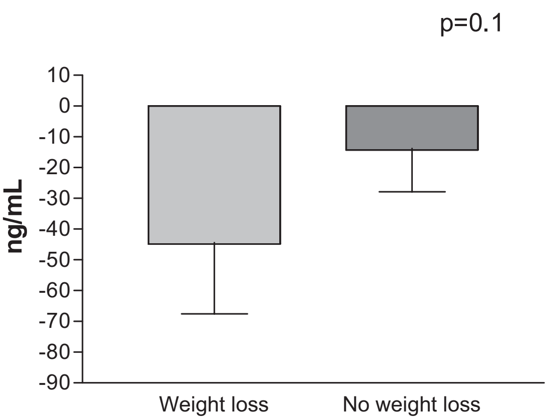

FIG. 2. A 24-wk change in PAI-1 for those completing the lifestyle treatment arm of the single treatment study separated by weight loss.

TABLE 4. Baseline characteristics of subjects in combination treatment trial

	PL (n = 18)	MET (n = 18)	*P*
Age (yr)	15.8 ± 1.6	14.7 ± 1.6	0.0422
BMI (kg/m^2)	35.7 ± 4.9	34.1 ± 4.3	ns
Waist (cm)	110.9 ± 8.4	109.7 ± 9.8	ns
Total T (ng/dl)	83.2 ± 27.8	102.6 ± 22.1	ns
SHBG (nmol/liter)	9.6 ± 11.9	10.1 ± 10.9	ns
FAI	6.3 ± 5.2	8.7 ± 5.9	ns
FG score	10.3 ± 4.3	7.9 ± 3.1	ns
Total CH (mg/dl)	160.1 ± 40.8	154.3 ± 41.2	ns
HDL (mg/dl)	36.9 ± 9.9	38.5 ± 10.4	ns
LDL (mg/dl)	99.1 ± 32.8	92.5 ± 33.3	ns
TG (mg/dl)	124.4 ± 82.9	103.3 ± 89.1	ns
Fasting ins (IU/ml)	52.9 ± 66.1	35.8 ± 39.1	ns
Fasting glu (mg/dl)	87.8 ± 15.1	88.5 ± 14.4	ns
SBP (mm Hg)	120.6 ± 15.3	121.2 ± 14.0	ns
DBP (mm Hg)	63.7 ± 9.5	66.9 ± 10.2	ns
CRP (mg/liter)	4.7 ± 5.5	4.6 ± 4.6	ns
AUC glu	18,049 ± 2,566	18,788 ± 2,948	ns
AUC ins	34,234 ± 20,739	30,763 ± 27,014	ns

PL, Placebo; MET, metformin; FG, Ferriman-Gallwey; SBP, systolic blood pressure; DBP, diastolic blood pressure; ns, not significant.

Lifestyle modification has been associated with improvements in endocrine and reproductive parameters in studies of adult women with PCOS (21, 42–44). Lifestyle modification alone was associated with significant compliance problems in the adolescent population in the first study. However, even with very modest weight change, subjects in the lifestyle treatment arm demonstrated improvements in SHBG and FAI, and this effect was more dramatic if weight loss was achieved. Improvements in PAI-1 and diastolic blood pressure were also noted.

Menstrual dysfunction and androgen excess in PCOS are typically managed by OC. Concurrent use of lifestyle treatment with OC may have the most benefit for those adolescent women with PCOS who are also obese. The role of metformin in this setting, however, remains unclear. The impact of metformin on testosterone suppression in our combined treatment trial, while statistically significant, is unlikely to produce a clinical improvement in androgen symptoms because free testosterone was also suppressed. This is likely mediated through increase in SHBG with OC, which was not enhanced by metformin in this study. Metformin did not enhance weight reduction in the lifestyle program in the combined treatment, but decreased central adiposity. Using a specific measure of visceral fat in the single treatment trial, subjects in the placebo arm significantly increased visceral adiposity in a 24-wk period. Central adiposity may be the most

TABLE 5. Baseline and 24-wk measures for subjects completing combination treatment trial

	PL (n = 16)		MET (n = 16)	
	Pre	Post	Pre	Post
BMI (kg/m^2)	35.1 ± 4.9	33.9 ± 4.2[b]	34.3 ± 4.6	32.4 ± 4.8[b]
Waist (cm)	111.1 ± 8.9	109.7 ± 8.4	110.1 ± 10.2	106.2 ± 11.7[a]
Total T (ng/dl)	83.3 ± 27.7	55.7 ± 32.5[a]	102.6 ± 22.1	44.9 ± 21.4[c,d]
SHBG (nmol/liter)	10.1 ± 12.5	68.1 ± 55.8[c]	9.5 ± 11.5	82.9 ± 52.4[c]
FAI	6.3 ± 5.2	1.1 ± 2.0[b]	8.7 ± 5.9	0.4 ± 0.6[c]
FG score	10.2 ± 4.6	7.0 ± 3.6[c]	8.4 ± 3.0	6.2 ± 1.9[c]
Total CH (mg/dl)	160.6 ± 43.3	182.0 ± 47.7[a]	160.4 ± 39.5	201.1 ± 42.9[b]
HDL (mg/dl)	37.2 ± 10.5	49.4 ± 13.1[c]	39.6 ± 10.6	57.9 ± 12.5[c,d]
LDL (mg/dl)	97.2 ± 34.4	104.4 ± 39.5	96.0 ± 33.8	115.4 ± 38.6[a]
TG (mg/dl)	132.4 ± 84.4	150.5 ± 112.8	110.9 ± 91.8	148.2 ± 64.3
FI (IU/ml)	52.9 ± 66.1	38.4 ± 37.4	36.6 ± 40.4	38.3 ± 39.5
AUC ins	34,234 ± 20,738	26,688 ± 9,761	31,325 ± 27,942	31,808 ± 37,371
AUC glu	18,104 ± 2,836	20,081 ± 9,761	18,973 ± 3,218	19,693 ± 3,222
FBS (mg/dl)	89.6 ± 17.5	82.2 ± 6.1	90.1 ± 15.4	81.6 ± 7.1
SBP (mm Hg)	118.8 ± 12.6	115.7 ± 13.1	118.6 ± 12.5	119.2 ± 13.3
DBP (mm Hg)	63.8 ± 10.1	62.1 ± 11.0	66.2 ± 10.4	63.1 ± 5.1
CRP (mg/liter)	4.2 ± 5.5	8.0 ± 19.3	4.1 ± 4.6	12.3 ± 21.5[a]
TG/HDL	3.8 ± 2.5	3.2 ± 2.6	3.4 ± 3.5	2.6 ± 1.6

PL, Placebo; MET, metformin; T, testosterone; FG, Ferriman-Gallwey; SBP, systolic blood pressure; DBP, diastolic blood pressure; FBS, fasting blood sugar; FI, fasting insulin; glu, glucose; ins, insulin.

[a] $P < 0.05$ compared to baseline.

[b] $P < 0.01$ compared to baseline.

[c] $P < 0.001$ compared to baseline.

[d] $P < 0.05$ compared to PL.

significant source of metabolic problems linked to obesity (45), and successful reduction in central adiposity may be linked to reduction in diabetes risk (46).

The current preliminary studies are limited by the small sample size and noncompliance with the lifestyle modification program in the first trial. Overall, however, these data are important due to the limited number of published trials of lifestyle management in this population. Difficulties with the implementation of the lifestyle program can be addressed in this population, as demonstrated by increased compliance in the second trial. Studies of adolescents with PCOS are also limited by the use of surrogate markers for cardiovascular outcomes. Based on studies in at-risk populations, the TG/HDL ratio is a good measure of insulin resistance and increased cardiovascular risk (47–49). Among subjects receiving lifestyle intervention plus OC, the TG/HDL ratio declined by 0.8 and 0.6 when metformin and placebo, respectively, were added (not significant). In future studies using this marker of cardiovascular risk, detection of a clinically significant difference of 0.4 in the reduction of TG/HDL ratio between treatment groups at the 5% significance level with 80% power will require 87 subjects per treatment arm. The results of these small randomized clinical trials suggest that there may be beneficial impact of a lifestyle program in combination with OC. Additionally, combining metformin with lifestyle modification and OC may enhance reduction of central adiposity, androgen suppression, and improvement in HDL. This remains to be confirmed in larger randomized trials. These studies represent short-term results, and it is not clear whether improvement in cardiovascular outcomes would result in the long term. The lessons of these small clinical trials of lifestyle management, either alone or in combination, will hopefully serve to stimulate additional randomized trials in the long-term management of PCOS in the obese adolescent.

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